

VC Dimension Upper-Bounds Rademacher Complexity For Binary Class Predictors

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Introduction

A central goal of statistical learning theory is to bound the generalisation error of a hypothesis class $\mathcal{F}$ uniformly across all data distributions. This poster traces a classical chain of results showing that, for binary classifiers, the Rademacher complexity, and consequently the worst-case generalisation gap, is governed by a single combinatorial parameter: the **Vapnik–Chervonenkis (VC) dimension**. The argument proceeds via Massart’s Finite Class Lemma, the growth function, and Sauer’s Lemma.

1. Motivation: The Generalisation Gap

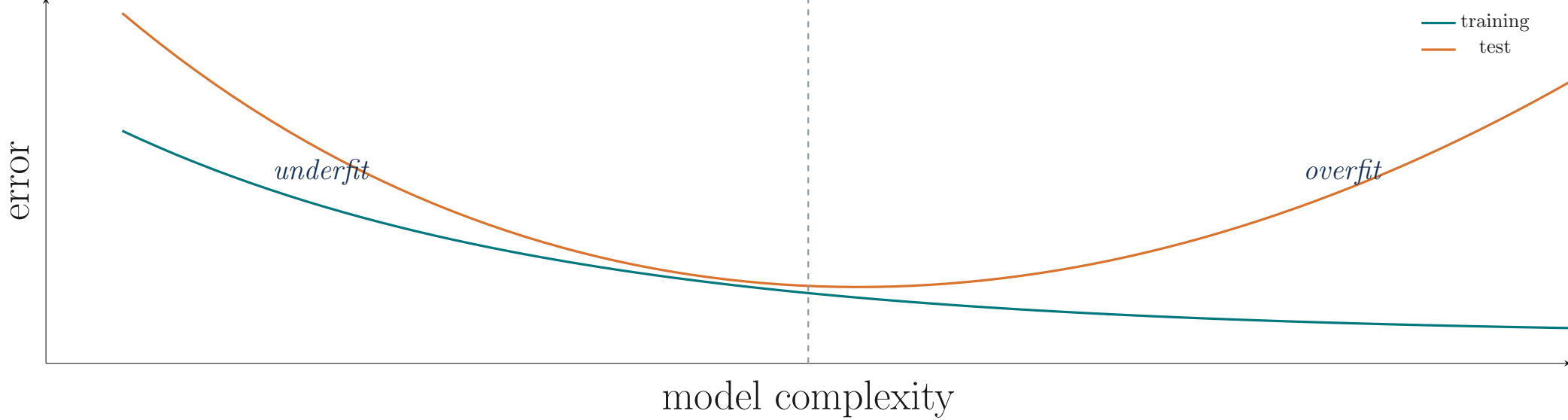
Statistical learning theory addresses the discrepancy between empirical risk, observed on training data, and population risk, which governs out-of-sample performance:

$$\widehat{R}(f) = \frac{1}{m} \sum_{i=1}^m \ell(y_i, f(x_i)), \quad R(f) = \mathbb{E}[\ell(y, f(x))].$$

The generalisation gap is $\Delta_f = \widehat{R}(f) - R(f)$. Since f is selected from $\mathcal{F}$ on the basis of the sample, the relevant quantity is the worst-case gap over the class:

$$\sup_{f \in \mathcal{F}} (\widehat{R}(f) - R(f)).$$

A uniform bound on this supremum underpins distribution-free generalisation guarantees.



The objective is to ensure empirical risk approximates true risk uniformly over $\mathcal{F}$.

2. Rademacher Complexity

Let $\epsilon_1, \dots, \epsilon_m$ be independent Rademacher variables with $\Pr[\epsilon_i = \pm 1] = 1/2$. The empirical Rademacher complexity of a class $\mathcal{H}$ on a sample $(z_1, \dots, z_m)$ is

$$\widehat{\mathfrak{R}}_m(\mathcal{H}) = \mathbb{E}_\epsilon \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \epsilon_i h(z_i) \right],$$

and $\mathfrak{R}_m(\mathcal{H}) = \mathbb{E}_S [\widehat{\mathfrak{R}}_m(\mathcal{H})]$ denotes its expectation over samples.

Symmetrisation. A standard argument yields

$$\mathbb{E}_S \left[\sup_{h \in \mathcal{H}} \left(\mathbb{E}[h(z)] - \frac{1}{m} \sum_{i=1}^m h(z_i) \right) \right] \leq 2 \mathfrak{R}_m(\mathcal{H}),$$

so a bound on $\mathfrak{R}_m(\mathcal{H})$ controls the expected uniform deviation of empirical from population means.

Interpretation. The Rademacher complexity quantifies the extent to which $\mathcal{H}$ correlates with random sign patterns: classes capable of fitting pure noise have high complexity, and consequently weak generalisation control (Bartlett).

3. From Functions to Label Vectors

For binary classifiers $\mathcal{F} \subseteq \{\pm 1\}^{\mathcal{X}}$, the Rademacher complexity depends on $\mathcal{F}$ only through its restriction to the sample:

$$\mathcal{F}|_{x_1^m} := \{(f(x_1), \dots, f(x_m)) : f \in \mathcal{F}\} \subseteq \{\pm 1\}^m.$$

Distinct functions may collapse to the same label vector; only $|\mathcal{F}|_{x_1^m}| \leq 2^m$ such vectors exist regardless of $|\mathcal{F}|$.

The **growth function** is the worst-case cardinality of this restriction:

$$\Pi_{\mathcal{F}}(m) = \max_{x_1, \dots, x_m \in \mathcal{X}} |\mathcal{F}|_{x_1^m}|.$$

From analysis to combinatorics. The complexity of an infinite class is reduced to a finite combinatorial quantity: the number of distinct label patterns realisable on m points. The trivial bound $\Pi_{\mathcal{F}}(m) \leq 2^m$ yields

$$\mathfrak{R}_m(\mathcal{F}) \leq \sqrt{\frac{2 \ln 2^m}{m}} = \sqrt{2 \ln 2},$$

which fails to vanish as $m \rightarrow \infty$. Establishing learnability therefore requires structural conditions ensuring $\Pi_{\mathcal{F}}(m) \ll 2^m$.

Capacity is measured by distinguishable behaviours on finite samples, not by the cardinality of $\mathcal{F}$.

4. Massart’s Finite Class Lemma

Theorem (Massart). Let $A \subset \mathbb{R}^m$ be finite and $r = \sup_{a \in A} \|a\|_2$. Then

$$\mathbb{E}_\epsilon \left[\sup_{a \in A} \frac{1}{m} \sum_{i=1}^m \epsilon_i a_i \right] \leq \frac{r \sqrt{2 \ln |A|}}{m}.$$

Proof sketch. With $\mu = \mathbb{E}_\epsilon [\sup_{a \in A} \sum_{i=1}^m \epsilon_i a_i]$, Jensen’s inequality and bounding the supremum by a sum yield, for any $\lambda > 0$,

$$e^{\lambda \mu} \leq \sum_{a \in A} \prod_{i=1}^m \mathbb{E}[e^{\lambda \epsilon_i a_i}].$$

Since $\mathbb{E}[e^{\lambda \epsilon_i a_i}] = \cosh(\lambda a_i) \leq e^{\lambda^2 a_i^2 / 2}$, this gives $e^{\lambda \mu} \leq |A| e^{\lambda^2 r^2 / 2}$. Minimising in λ produces the stated bound.

Application to $\mathcal{F}$. Setting $A = \mathcal{F}|_{x_1^m}$ and noting $\|a\|_2 = \sqrt{m}$ for any $a \in \{\pm 1\}^m$:

$$\mathfrak{R}_m(\mathcal{F}) \leq \frac{\sqrt{m} \cdot \sqrt{2 \ln \Pi_{\mathcal{F}}(m)}}{m} = \sqrt{\frac{2 \ln \Pi_{\mathcal{F}}(m)}{m}}.$$

Critical points.

- Sub-Gaussian concentration.** The bound $\cosh(\lambda a) \leq e^{\lambda^2 a^2 / 2}$ encodes that ϵa is sub-Gaussian with variance proxy a^2 ; this is the same machinery underlying Hoeffding’s inequality.
- Origin of the $\sqrt{\log |A|}$ factor.** It arises from replacing the supremum with a sum, effectively a union bound. Tightening it requires chaining.
- Universality.** The lemma applies to any finite subset of $\mathbb{R}^m$; its application to function classes is one instance of a general tool.

5. VC Dimension

A set $\{x_1, \dots, x_m\}$ is **shattered** by $\mathcal{F}$ if

$$\mathcal{F}|_{x_1^m} = \{\pm 1\}^m,$$

that is, every binary labelling is realised by some $f \in \mathcal{F}$. The Vapnik–Chervonenkis dimension is the supremum of cardinalities shattered by $\mathcal{F}$:

$$\text{VCdim}(\mathcal{F}) = \sup\{m : \Pi_{\mathcal{F}}(m) = 2^m\}.$$

$\mathcal{X}$	$\mathcal{F}$	$\text{VCdim}(\mathcal{F})$
$\mathbb{R}^2$	convex polygons	∞
$\mathbb{R}^2$	axis-aligned rectangles	4
$\mathbb{R}^2$	convex polygons with d vertices	$2d + 1$
$\mathbb{R}^d$	halfspaces	$d + 1$

Halfspaces in $\mathbb{R}^d$. The lower bound is achieved by shattering $\{0, e_1, \dots, e_d\}$ via suitable choices of weights and offset. The upper bound follows from **Radon’s Lemma**: any $d + 2$ points in $\mathbb{R}^d$ admit a partition X_1, X_2 with $\text{conv}(X_1) \cap \text{conv}(X_2) \neq \emptyset$. The labelling assigning $+1$ to X_1 and -1 to X_2 then cannot be realised by any halfspace, since a halfspace assigning a label to a set assigns the same label to its convex hull.

6. Sauer’s Lemma

Theorem (Sauer). If $\text{VCdim}(\mathcal{F}) \leq d$, then

$$\Pi_{\mathcal{F}}(m) \leq \sum_{i=0}^d \binom{m}{i},$$

and for $m \geq d$,

$$\Pi_{\mathcal{F}}(m) \leq \left(\frac{em}{d} \right)^d.$$

Proof idea. The proof uses induction on $m + d$. Removing one point splits the labellings into:

- labellings with a unique extension,
- labellings admitting both ± 1 extensions.

If a subset of labellings admits both extensions, its effective VC dimension drops by one:

$$\text{VCdim}(\mathcal{F}_3) \leq d - 1.$$

Applying the inductive hypothesis recursively forces polynomial growth:

$$2^m \longrightarrow m^d.$$

Critical insights.

- Combinatorial collapse.** Failure to shatter beyond d points forces polynomial rather than exponential growth.
- Purely combinatorial.** The proof uses counting and shattering structure rather than probability or geometry.
- Historical importance.** Sauer’s lemma is one of the foundational links between combinatorics and statistical learning theory.

7. From VC Dimension to Generalisation

Combining Massart’s Lemma with Sauer’s Lemma, for any binary class $\mathcal{F}$ with $\text{VCdim}(\mathcal{F}) \leq d$:

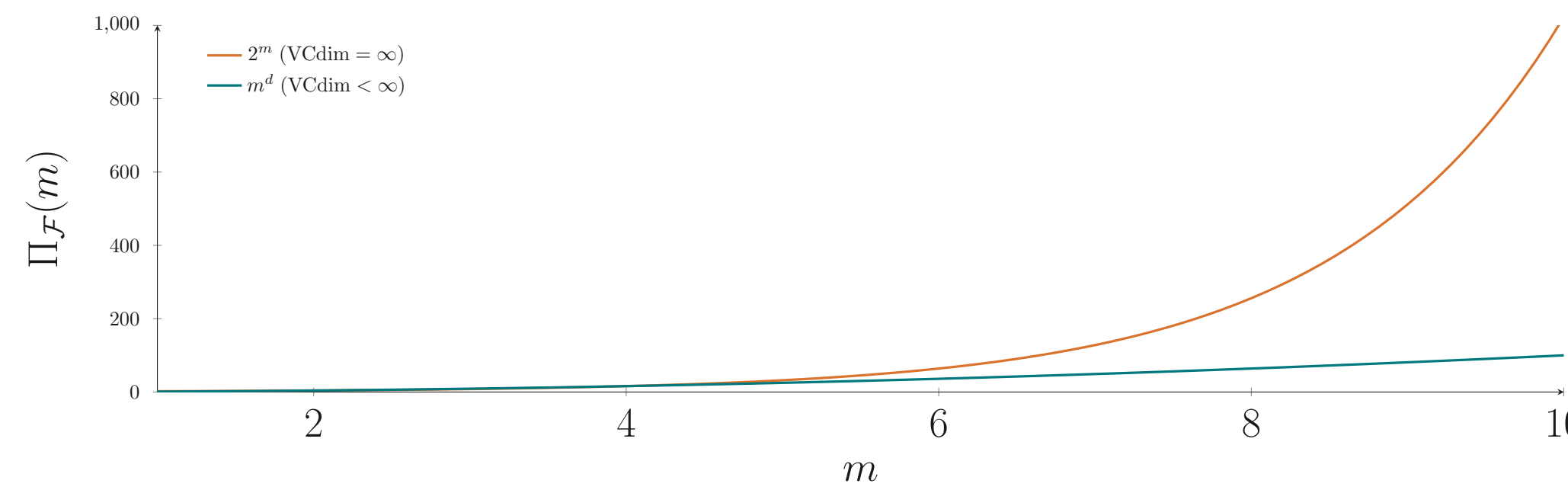
$$\mathfrak{R}_m(\mathcal{F}) \leq \sqrt{\frac{2 \ln \Pi_{\mathcal{F}}(m)}{m}} \leq \sqrt{\frac{2d \ln(em/d)}{m}}.$$

Dichotomy of growth. If $\text{VCdim}(\mathcal{F}) = \infty$, then $\Pi_{\mathcal{F}}(m) = 2^m$ for every m . Otherwise, Sauer’s Lemma implies polynomial growth:

$$\Pi_{\mathcal{F}}(m) = O(m^d).$$

No intermediate growth rate occurs.

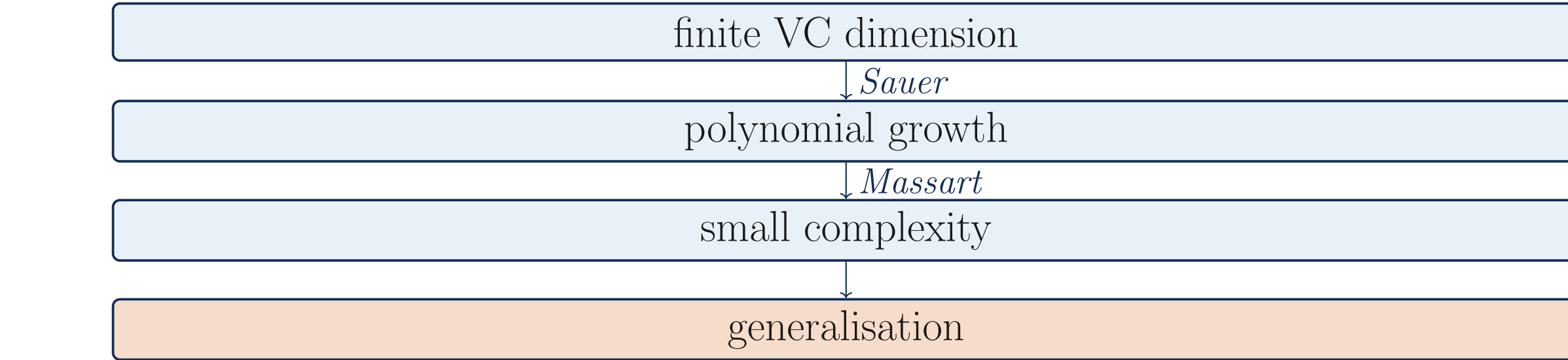
Interpretation. Finite VC dimension prevents the hypothesis class from sustaining exponential combinatorial growth.



Consequence. For $\text{VCdim}(\mathcal{F}) = d < \infty$,

$$\mathfrak{R}_m(\mathcal{F}) = O\left(\sqrt{\frac{d \ln m}{m}}\right) \rightarrow 0 \quad \text{as } m \rightarrow \infty.$$

Consequently, the expected gap between empirical and true performance vanishes with more data.



Finite VC dimension, a purely combinatorial property, is sufficient to guarantee learnability independently of the underlying data distribution.

8. Discussion

Tightness. The $\sqrt{\log m/m}$ rate retains a logarithmic factor that more refined techniques can remove. Chaining and Dudley’s entropy integral yield bounds of the form $\mathfrak{R}_m(\mathcal{F}) = O(\sqrt{d/m})$ without the log factor. The bound is also distribution-free; data-dependent variants based on the empirical Rademacher complexity are typically tighter in practice.

Scope. The reduction relies on $\|a\|_2 = \sqrt{m}$, which exploits the binary structure $\mathcal{F} \subseteq \{\pm 1\}^{\mathcal{X}}$. Real-valued classes require alternative complexity measures: the *pseudo-dimension* and the *fat-shattering dimension* generalise the framework to such settings.

The dichotomy. Sauer’s Lemma reveals that growth functions exhibit only two regimes: exponential when $\text{VCdim}(\mathcal{F}) = \infty$, and eventually polynomial when $\text{VCdim}(\mathcal{F}) < \infty$. No intermediate growth rate is possible, a structurally striking feature of binary hypothesis classes.

Connection to PAC learning. The fundamental theorem of statistical learning establishes that a class is PAC-learnable if and only if it has finite VC dimension, with sample complexity $O(d/\varepsilon^2 \cdot \log(1/\delta))$. VC theory thus characterises learnability itself, not merely a sufficient condition.

Modern caveat. Deep neural networks possess effectively unbounded VC dimension yet generalise well in practice. Classical VC theory does not predict this behaviour, motivating margin-based bounds, PAC-Bayes analysis (Dziugaite & Roy, 2017), and approaches based on implicit regularisation and the neural tangent kernel.

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